

Question description

Byteland has n cities and m roads between them. The goal is to construct new roads so that there is a route between any two cities. Your task is to find out the minimum number of roads required, and also determine which roads should be built.

Constraints

$$1 \leq n \leq 10^5$$

$$1 \leq m \leq 2 \cdot 10^5$$

$$1 \leq a, b \leq n$$

Input

The first input line has two integers n and m : the number of cities and roads. The cities are numbered $1, 2, \dots, n$.

After that, there are m lines describing the roads. Each line has two integers a and b : there is a road between those cities.

A road always connects two different cities, and there is at most one road between any two cities.

Output

First print an integer k : the number of required roads.

Then, print k lines that describe the new roads. You can print any valid solution.

✓ Logical Test Cases

Test Case 1
INPUT (STDIN)
4 2 1 2 3 4
EXPECTED OUTPUT
1 2 4

Test Case 2
INPUT (STDIN)
4 2 2 3 2 4
EXPECTED OUTPUT
1 1 4

✓ Mandatory Test Cases

Test Case 1
KEYWORD
<code>while(m--)</code>

✓ Complexity Test Cases

Test Case 1
CYCLOMATIC COMPLEXITY
5

Test Case 2
TOKEN COUNT
475

Test Case 3
NLOC
50

Graph4.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAXN 100005
5
6 int parent[MAXN];
7
8 int find(int i) {
9     if (parent[i] == i)
10         return i;
11     return parent[i] = find(parent[i]);
12 }
13
14 int main() {
15     int n, m;
16     if (scanf("%d %d", &n, &m) != 2) return 0;
17
18     for (int i = 1; i <= n; i++) {
19         parent[i] = i;
20     }
21
22     while (m--) {
23         int u, v;
24         scanf("%d %d", &u, &v);
25         int fu = find(u),
```

```
26     int rv = find(v);
27     if (ru != rv) {
28         parent[ru] = rv;
29     }
30 }
31
32 int reps[MAXN];
33 int rep_cnt = 0;
34 for (int i = 1; i <= n; i++) {
35     if (find(i) == i) {
36         reps[rep_cnt++] = i;
```

```
37     }
38 }
39
40 printf("%d\n", rep_cnt - 1);
41 for (int i = 1; i < rep_cnt; i++) {
42     printf("%d %d\n", reps[i - 1], reps[i]);
43 }
44
45 return 0;
46}
47
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

4 2

1 2

3 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Graph4.exe

1

2 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

4 2

2 3

2 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Graph4.exe

1

1 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>